Global Air Pollution -Exploratory Data Analysis and Variable Selection

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying statistical or machine learning models. It helps identify the structure, distribution, relationships, missing values, duplicates, and unusual observations present in the data.

In this project, EDA is particularly important because we do not want to select the independent variable arbitrarily. We have multiple pollutant-related numerical variables, so EDA will help us compare them and determine which one has the strongest relationship with AQI Value.

EDA is performed to:

- Understand the structure and characteristics of the dataset.
- Identify numerical and categorical variables.
- Check for missing and duplicate values.
- Understand the distribution of numerical variables.
- Detect potential outliers.
- Identify relationships between variables.
- Support data-driven feature/variable selection.
- Determine whether a variable is suitable for a particular machine learning model.

Visualization converts numerical information into graphical form, making patterns and relationships easier to identify. It allows us to visually examine distributions, outliers, trends, and relationships that may not be immediately apparent from numerical values alone.

In this project, visualization will be used to:

- Examine the distribution of each numerical variable using histograms.
- Identify potential outliers using boxplots.
- Examine the relationship between each pollutant and AQI Value using scatter plots.
- Compare the strength of relationships using correlation analysis and a heatmap.

Step 1: importing libraries

The required Python libraries were imported for data manipulation, numerical computations, and visualization. Pandas was used for handling the dataset, NumPy for numerical operations, and Matplotlib and Seaborn for statistical visualization.

Step 2:

After importing the required libraries, the next step is to load the Global Air Pollution Dataset into Python. Since the dataset is stored in CSV format, we use Pandas' read_csv() function.

The data is stored in a Pandas DataFrame, which allows us to work with the dataset in a structured rows-and-columns format.

Step 3: Understanding the Dataset

Before selecting a predictor, we need to understand the structure and quality of the dataset. This helps us identify the available variables, their data types, the size of the dataset, and whether missing or duplicate values are present.

- df.shape → Returns the number of rows and columns.
- df.dtypes → Displays the data type of each column.
- df.describe() → Provides statistical summaries of numerical columns.
- df.isnull().sum() → Counts missing values in each column.

Step 4:Histogram plots for distribution

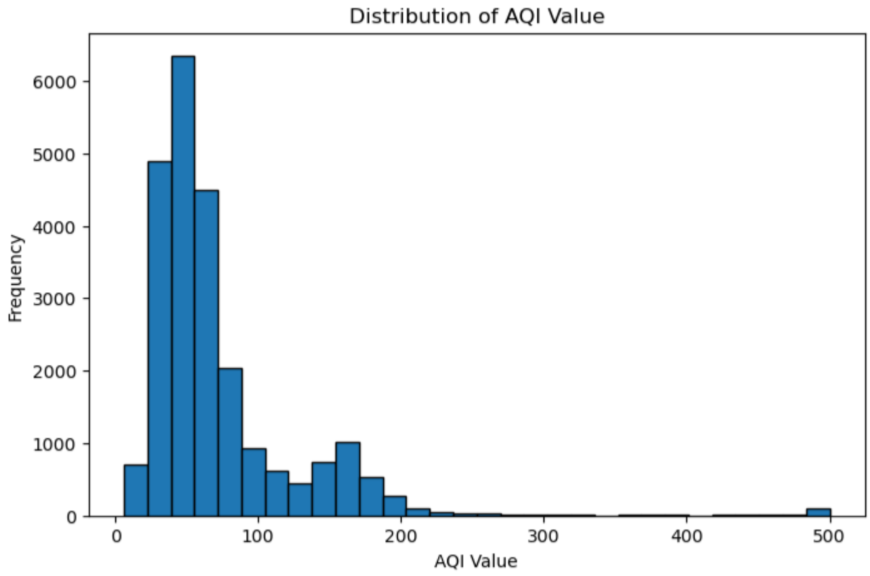
A histogram is used to understand the distribution of a numerical variable. It shows how frequently values occur within different ranges and helps us observe the spread, concentration, and skewness of the data.

In this project, histograms will be created for all five numerical variables to understand their individual distributions before selecting a predictor.

- plt.hist() → Creates a histogram to show the distribution of values.
- bins=30 → Divides the data into 30 intervals.
- plt.figure() → Sets the size of the plot.
- plt.xlabel() → Labels the x-axis.
- plt.ylabel() → Labels the y-axis.
- plt.title() → Adds a title to the plot.
- plt.show() → Displays the plot.

for column in numeric_columns → Repeats the process for every numerical variable.

Plot 1: Distribution of AQI Values



The AQI Value distribution is concentrated at relatively lower values and is positively skewed, with a long tail toward higher AQI values. A small number of observations have very high AQI values.

If a variable is **right-skewed (positively skewed)**, most of its values are concentrated on the **left/lower side**, while a few unusually large values stretch the distribution toward the **right**.

For a right-skewed variable:

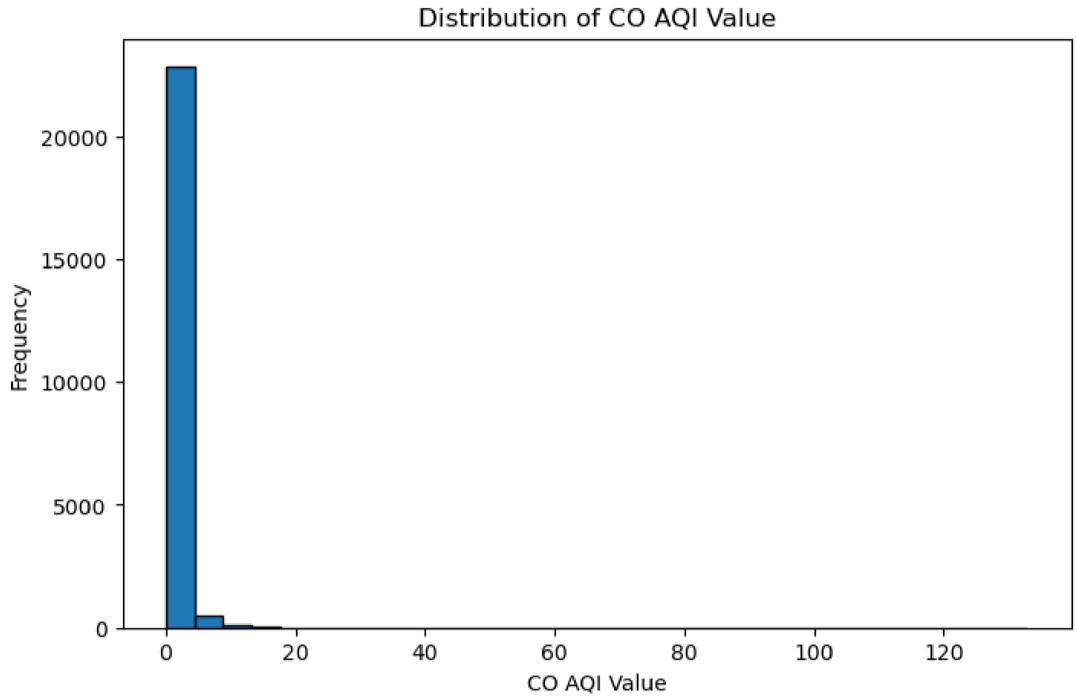
Mode < Median < Mean

In data analysis

If a variable is strongly right-skewed:

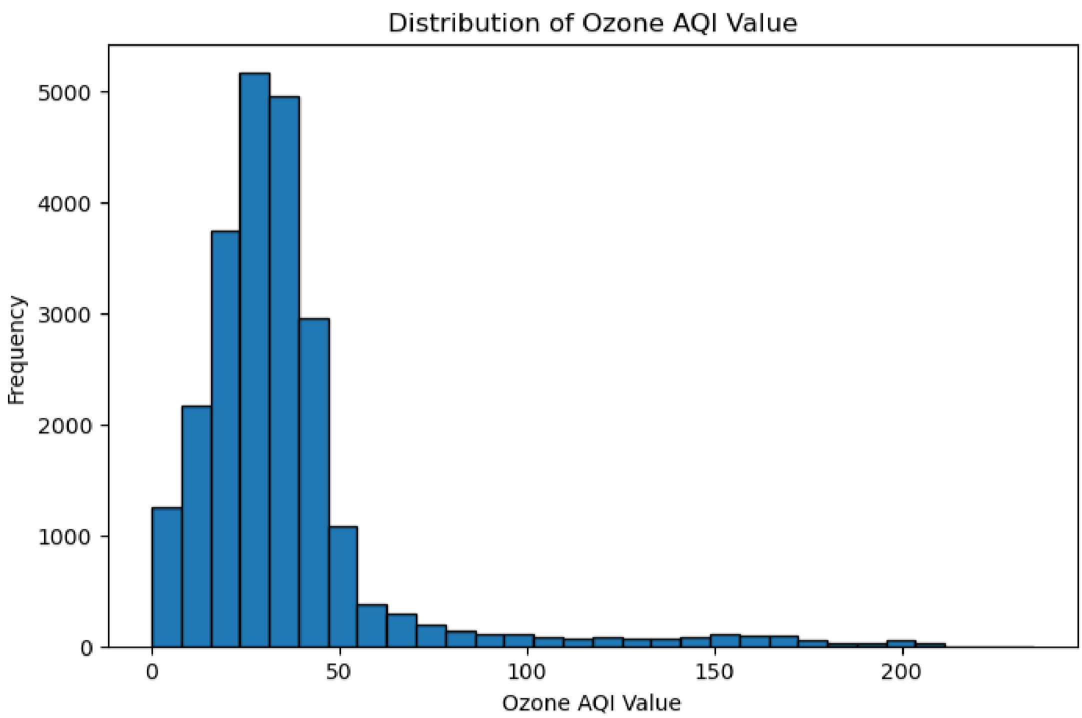
- The **mean may not represent the typical value well** because it is affected by large outliers.
- The **median is often a better measure of central tendency**.
- You may consider a **log transformation** or another transformation before applying some statistical/ML methods.

Plot 2: CO AQI Value distribution



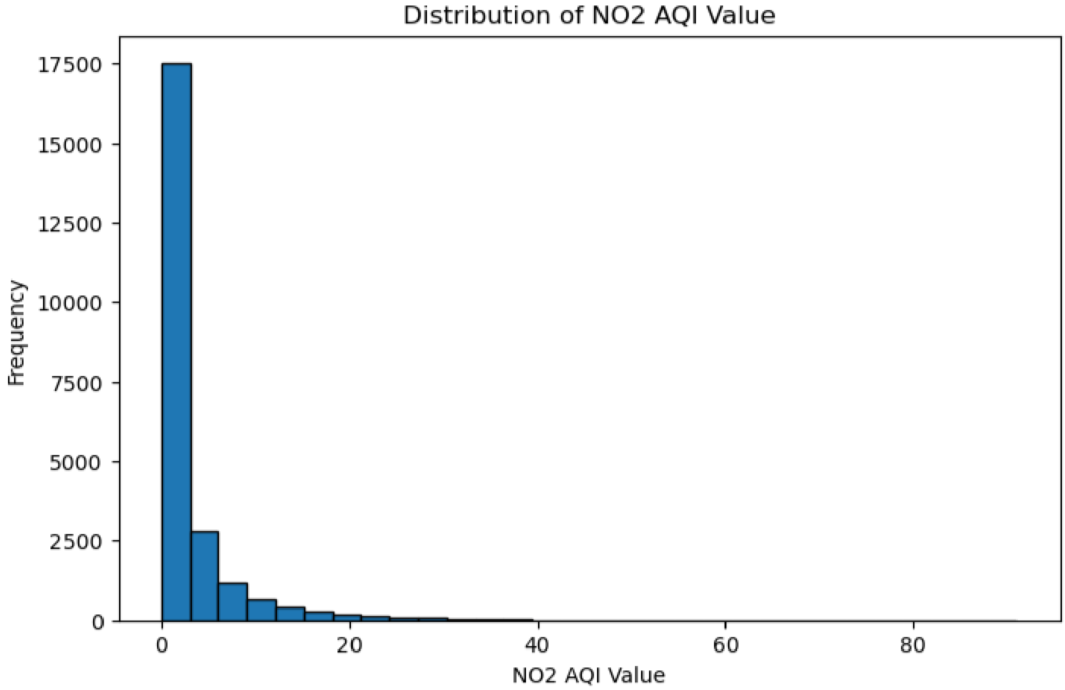
This distribution is highly concentrated near the lower values, particularly around 0–5. The frequency decreases rapidly as the CO AQI Value increases. A long right tail extends toward higher values, reaching approximately 133.

Plot 3: Ozone AQI Value



Ozone values are mainly concentrated around approximately 20–50. The frequency decreases as the values increase, producing a noticeable right-skewed distribution. Some observations extend to values above 200.

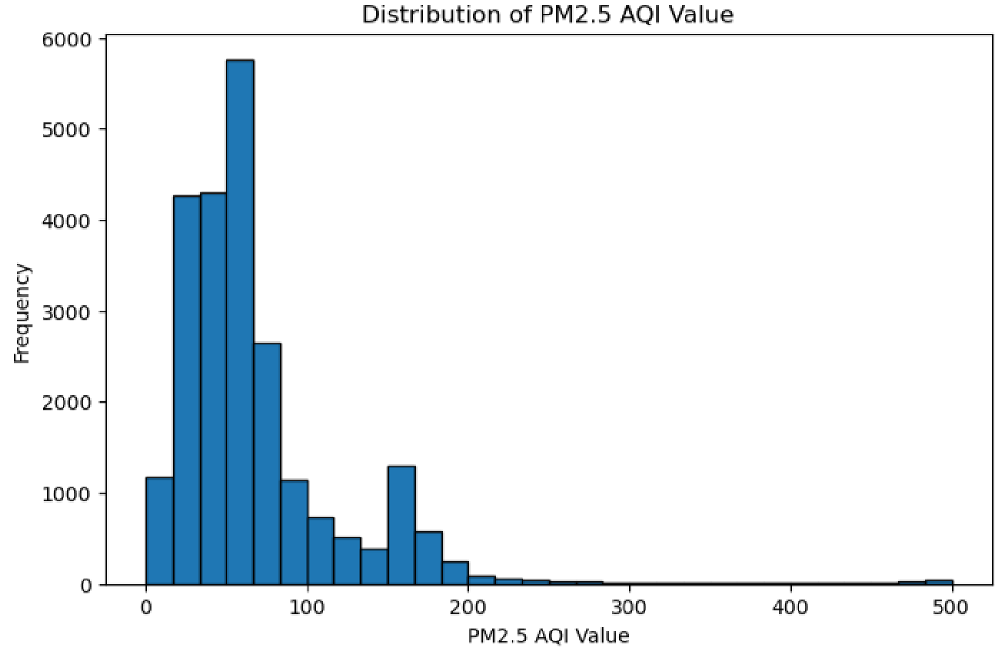
Plot 4: NO2 AQI Value distribution



The NO2 AQI Value distribution is highly concentrated at low values, with most observations occurring between approximately 0 and 10. The frequency decreases as the NO2 AQI Value increases, with the main distribution extending up to around 40.

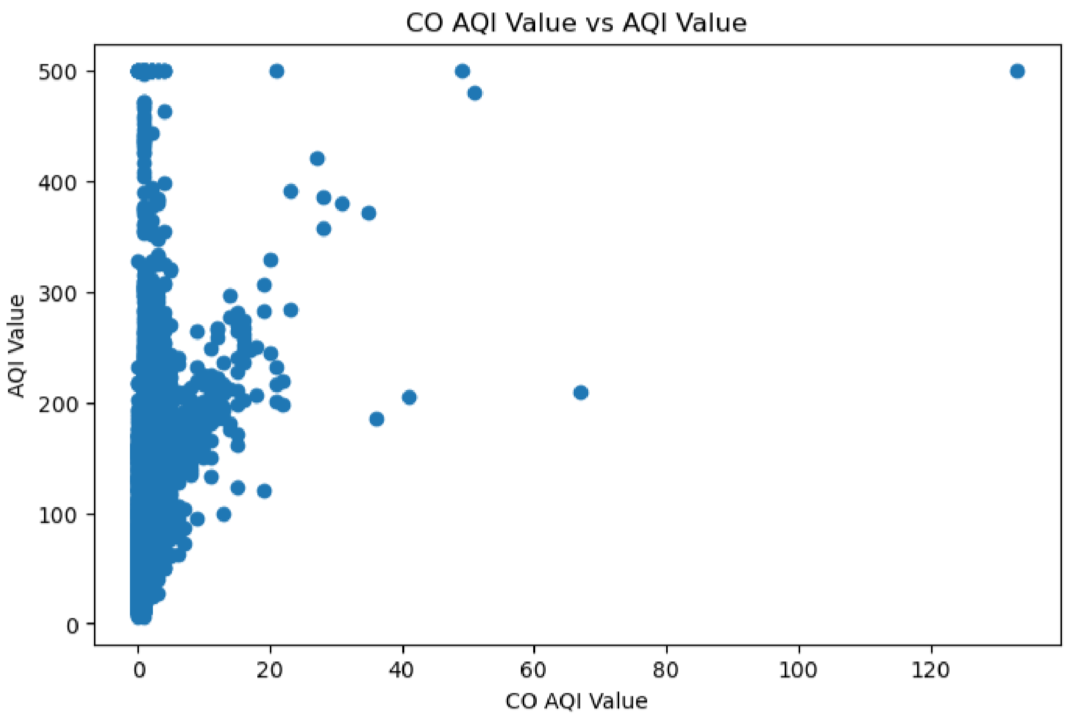
Plot 5: PM2.5 AQI Value distribution

NO2 AQI Value



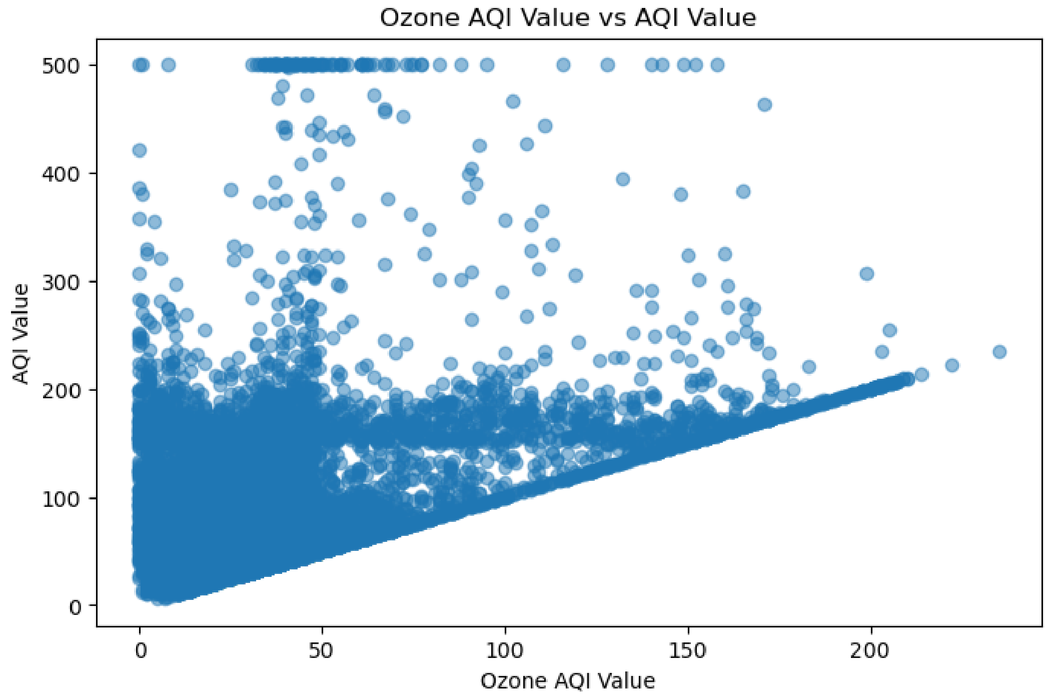
Most PM2.5 values are concentrated roughly between 20 and 100, with a peak around 50–70. There is a noticeable right tail extending toward 500, including a small number of very high observations.

Plot 6: CO AQI Value vs AQI Value



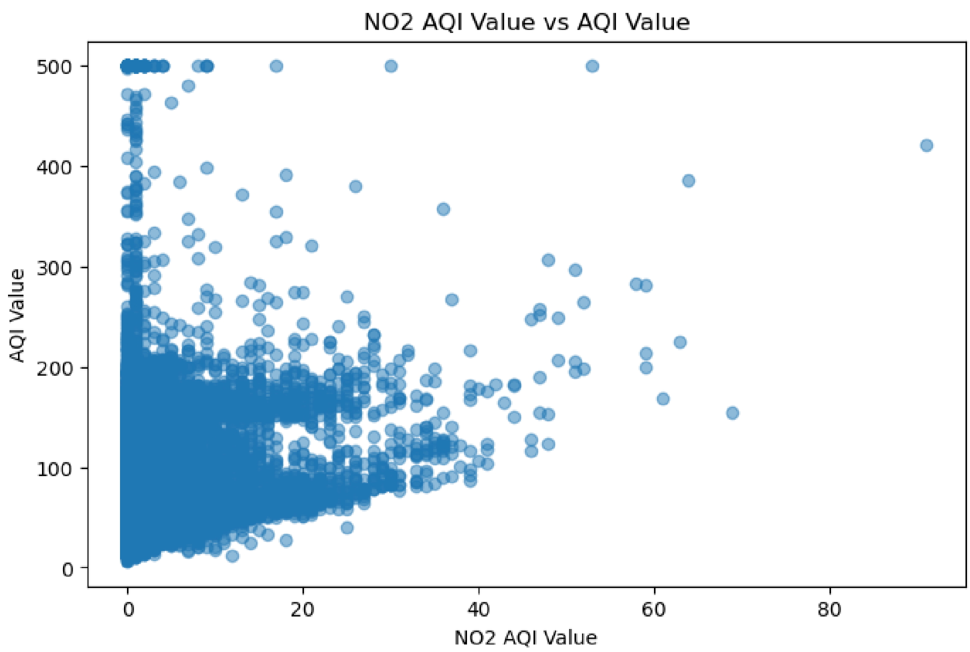
The plot shows a positive relationship, but the observations are highly scattered. Most CO values are concentrated near the lower end, while AQI values vary considerably for similar CO values.

Plot 7: Ozone AQI Value vs AQI Value



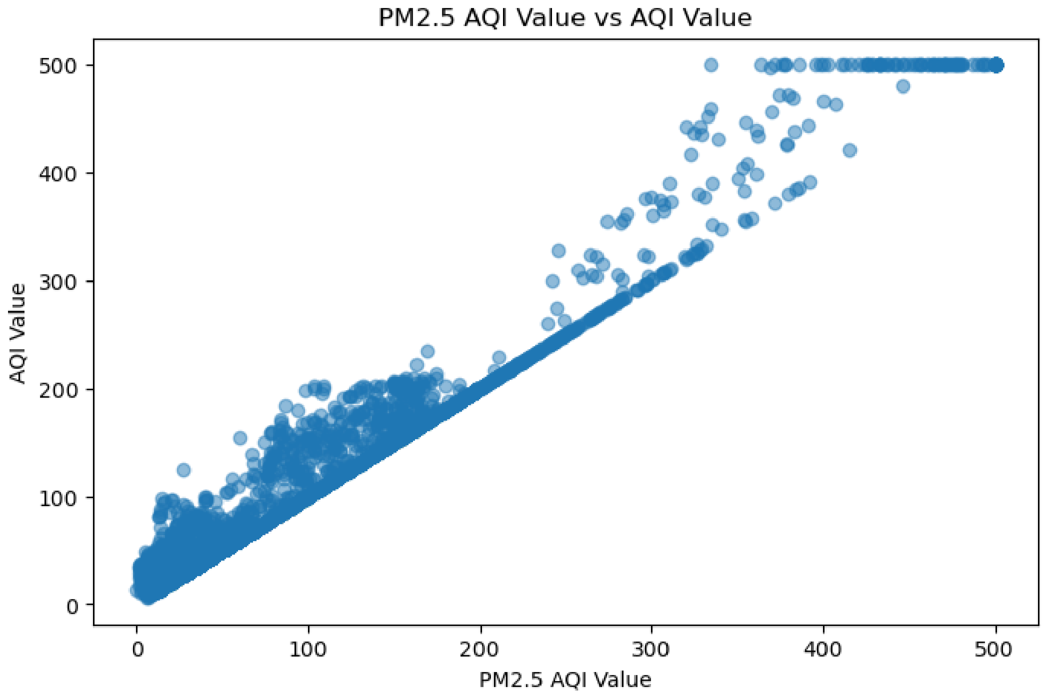
The scatter plot shows a positive relationship between Ozone AQI Value and AQI Value, with some outliers present in the data.

Plot 8: NO2 AQI Value vs AQI Value



Most NO2 observations are concentrated at low values, while AQI Value varies substantially within this range. The overall pattern is relatively scattered

Plot 9: PM2.5 AQI Value vs AQI Value



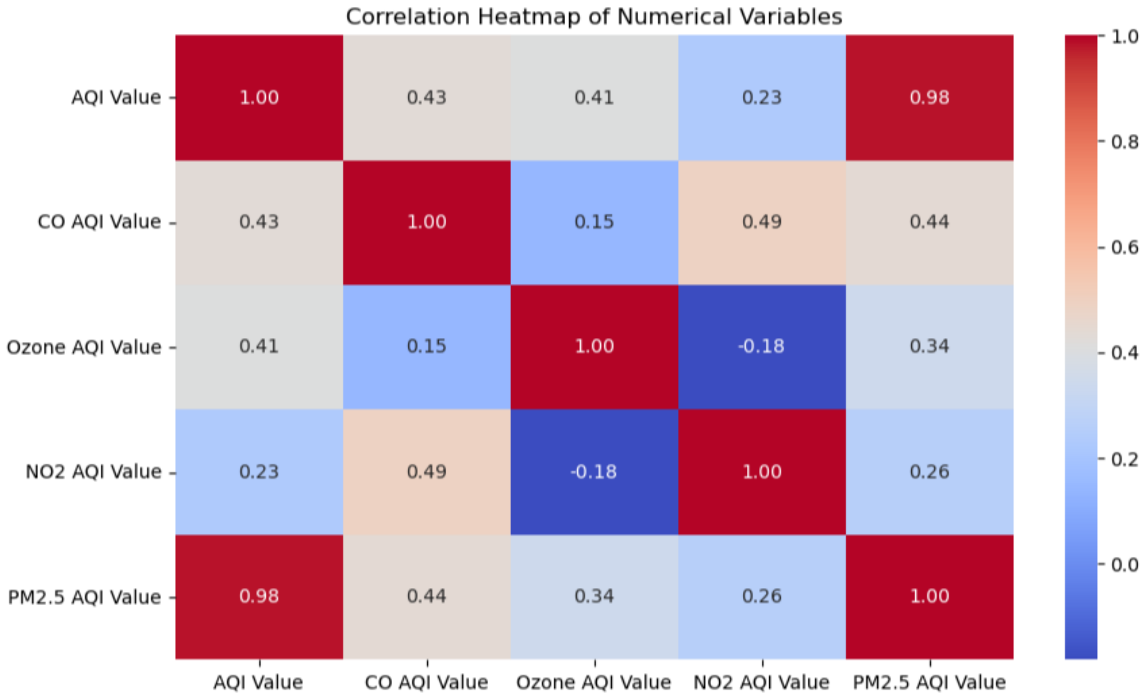
This is clearly the strongest relationship among the four. The observations follow a very clear upward pattern, with AQI Value increasing as PM2.5 AQI Value increases. The points are much more closely aligned with a linear trend.

Correlation analysis

Correlation measures the strength and direction of the linear relationship between two numerical variables. The correlation coefficient ranges from -1 to +1, where values closer to +1 indicate a strong positive linear relationship.

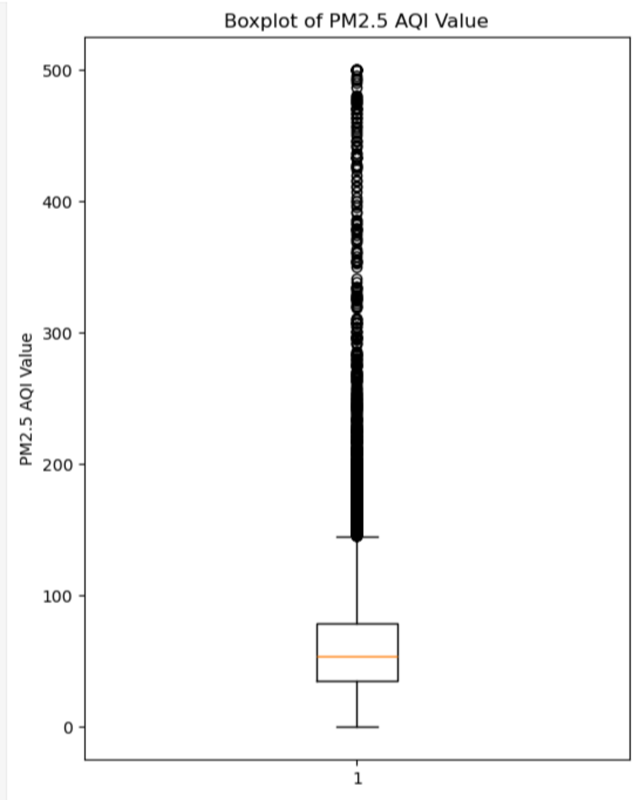
- CO AQI Value → 0.430602: moderate positive relationship.
- Ozone AQI Value → 0.405310: moderate positive relationship.
- NO2 AQI Value → 0.231758: weak positive relationship.
- PM2.5 AQI Value → 0.984327: extremely strong positive linear relationship.

Plot 10: Correlation Heatmap



The heatmap shows that PM2.5 AQI Value has the strongest positive correlation with AQI Value, with a correlation coefficient of 0.98. CO and Ozone show moderate positive correlations of 0.43 and 0.41 respectively, while NO2 has a weaker positive correlation of 0.23.

After selection of variable



These potential outliers will not be removed at this stage because they are part of the original dataset. During the improved regression stage, an appropriate outlier-treatment technique will be applied, and the model will be retrained to investigate whether removing the extreme observations improves its performance.